

AI CAREER FOR WOMEN

Engineering Spoke



Project Title Name
AI-Powered Crop Recommendation System Using Machine Learning

A Project Report

submitted in partial fulfillment of the requirements
of
AICW project.by

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Under the Guidance of
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ACKNOWLEDGEMENT

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We sincerely thank everyone who directly or indirectly contributed to the successful completion of this project.

Team Members

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ABSTRACT of the Project

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Agriculture plays a crucial role in the economic development and food security of a nation. Selecting the most suitable crop for cultivation is a challenging task for farmers due to variations in soil nutrients and environmental conditions. Incorrect crop selection can lead to reduced productivity and financial losses. To address this issue, this project proposes an AI-Powered Crop Recommendation System using Machine Learning techniques.

The main objective of the project is to recommend the most suitable crop based on soil and climatic parameters such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH value, and rainfall. The system utilizes a Crop Recommendation Dataset containing 2200 records and applies data preprocessing, feature analysis, model training, and prediction techniques to generate accurate recommendations.

A Random Forest Classifier is used as the machine learning algorithm due to its high accuracy and reliability in handling agricultural data. The dataset is divided into training and testing sets, and the model is trained to identify patterns between environmental conditions and crop types. Various data visualization techniques, including crop distribution analysis and correlation heatmaps, are used to better understand the dataset.

The developed model achieved an accuracy of approximately 99.32%, demonstrating excellent prediction performance. Based on user-provided soil and environmental parameters, the system successfully recommends the most appropriate crop for cultivation.

The proposed system can assist farmers in making informed decisions, improving agricultural productivity, reducing crop failure risks, and promoting efficient utilization of available resources. This project highlights the potential of Artificial Intelligence and Machine Learning in supporting modern agricultural practices and sustainable farming.

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CHAPTER 1

Introduction

1.1 Problem Statement: Agriculture is one of the most important sectors that contributes significantly to food production and economic development. Farmers often face difficulties in selecting the most suitable crop for cultivation because of variations in soil nutrients and environmental conditions. Traditional methods of crop selection mainly depend on experience and assumptions, which may lead to poor crop yield, inefficient use of resources, and financial losses. Therefore, there is a need for an intelligent system that can analyze soil and climatic parameters and recommend the most appropriate crop. This project addresses this problem by developing a Machine Learning-based Crop Recommendation System that helps farmers make accurate and data-driven decisions.

1.2 Motivation: The rapid growth of Artificial Intelligence and Machine Learning has created opportunities to solve real-world agricultural problems effectively. Agriculture remains the primary source of livelihood for millions of people, and improving crop productivity is essential for sustainable development. This project was chosen to assist farmers in selecting suitable crops based on scientific analysis rather than traditional methods. The system can reduce crop failure risks, improve productivity, increase profitability, and support precision farming practices. It can also be integrated into agricultural advisory systems and smart farming applications.

1.3 Objective: The main objectives of this project are:

- 1.To analyze soil nutrient and environmental data.
- 2.To preprocess and prepare agricultural datasets for machine learning.
- 3.To develop a crop recommendation model using the Random Forest Classifier algorithm.
- 4.To predict and recommend the most suitable crop based on user inputs.
- 5.To improve agricultural decision-making and crop productivity.
- 6.To achieve high prediction accuracy using machine learning techniques.

1.4 Scope of the Project: The scope of this project is limited to recommending crops based on soil nutrients (Nitrogen, Phosphorus, Potassium) and environmental parameters such as temperature, humidity, pH, and rainfall. The system uses a machine learning model trained on historical agricultural data to provide crop recommendations.

Limitations:

- 1.The accuracy of recommendations depends on the quality and correctness of input data.
- 2.The system is trained on a specific dataset and may require retraining for different regions.
- 3.Market demand, pest attacks, and government policies are not considered in the recommendation process.
- 4.Real-time weather forecasting is not integrated into the current system.

CHAPTER 2

Literature Survey

2.1 Review relevant literature or previous work in this domain.

Review of Relevant Literature

Agriculture is a critical sector that directly influences food security and economic growth. In recent years, researchers have focused on applying Artificial Intelligence (AI) and Machine Learning (ML) techniques to improve agricultural productivity and decision-making. Several studies have proposed crop recommendation systems that analyze soil and environmental conditions to identify suitable crops for cultivation.

Various machine learning algorithms such as Decision Trees, Naïve Bayes, Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Random Forest have been used in agricultural applications. These systems utilize factors such as soil nutrients, temperature, humidity, rainfall, and pH values to provide crop recommendations. The use of data-driven approaches has shown significant improvements in prediction accuracy compared to traditional farming practices.

2.2 Existing Models, Techniques, and Methodologies

Several crop recommendation systems have been developed using machine learning and data mining techniques. Commonly used methodologies include:

Decision Tree Algorithm – Used for classification and crop prediction based on agricultural parameters.

Naïve Bayes Classifier – Provides probabilistic crop recommendations using statistical analysis.

Support Vector Machine (SVM) – Used for classification tasks with high-dimensional agricultural datasets.

K-Nearest Neighbors (KNN) – Predicts suitable crops based on similarities between data points.

Random Forest Classifier – An ensemble learning technique that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Among these techniques, Random Forest has demonstrated better performance in handling large agricultural datasets and producing highly accurate recommendations.

2.3 Gaps and Limitations in Existing Solutions

Although several crop recommendation systems exist, many of them have certain limitations:

Some systems provide low prediction accuracy due to insufficient training data.

Many models consider only a limited number of soil parameters.

Existing solutions may not effectively handle large and complex datasets.

Traditional methods often rely on manual analysis and expert knowledge.

Some systems lack user-friendly interfaces for farmers.

To overcome these limitations, the proposed project uses the Random Forest Classifier with a comprehensive dataset containing soil nutrients and environmental parameters. The system achieves high prediction accuracy and provides reliable crop recommendations. By utilizing

machine learning techniques, the proposed solution supports data-driven agricultural decision-making and helps farmers improve productivity and reduce crop selection risks.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 System Design

The proposed Crop Recommendation System is designed to recommend the most suitable crop based on soil nutrients and environmental conditions. The system uses machine learning techniques to analyze user inputs and predict the best crop for cultivation.

3.1.1 Data Collection

The Crop Recommendation Dataset containing 2200 records was collected. The dataset includes Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, Rainfall, and Crop Label.

3.1.2 Data Preprocessing

The collected dataset was cleaned and analyzed to remove inconsistencies. Data preprocessing techniques were applied to improve model performance and ensure accurate predictions.

3.1.3 Model Training

The Random Forest Classifier algorithm was used to train the machine learning model. The dataset was divided into training and testing datasets.

3.1.4 Crop Prediction

After training, the model predicts the most suitable crop based on user-provided soil and environmental parameters.

3.2 Modules Used

3.2.1 Dataset Management Module

Responsible for loading, organizing, and preparing agricultural data.

3.2.2 Data Visualization Module

Used for generating crop distribution graphs and correlation heatmaps.

3.2.3 Machine Learning Module

Implements the Random Forest Classifier for crop recommendation.

3.2.4 Prediction Module

Accepts user inputs and generates crop recommendations.

3.3 Data Flow Diagram

3.3.1 DFD Level 0

User → Input Soil & Climate Data → Crop Recommendation System → Recommended Crop

3.3.2 DFD Level 1 – Data Collection Module

Dataset → Data Collection → Dataset Storage

3.3.3 DFD Level 1 – Model Training Module

Dataset → Preprocessing → Model Training → Trained Model

3.3.4 DFD Level 1 – Crop Prediction Module

User Input → Trained Model → Crop Recommendation Output

3.4 Advantages

1. High prediction accuracy.
2. Supports data-driven farming decisions.
3. Reduces crop selection risks.
4. Improves agricultural productivity.
5. Easy to use and cost-effective.

3.5 Requirement Specification

3.5.1 Hardware Requirements

- Laptop/Desktop Computer

- Intel i3 Processor or Higher
- Minimum 4 GB RAM
- 10 GB Free Storage

Software Requirements

- Windows 10/11
 - Python 3.x
 - Google Colab / Jupyter Notebook
 - Pandas
 - NumPy
 - Scikit-learn
 - Matplotlib
 - GitHub

CHAPTER 4

IMPLEMENTATION AND RESULTS

4.1 Dataset Analysis Results

The Crop Recommendation Dataset contains 2200 records and 8 attributes. Exploratory Data Analysis was performed to understand the distribution of crops and feature relationships.

4.2 Results of Model Training

The Random Forest Classifier was trained using the agricultural dataset. The dataset was split into training and testing datasets. The model learned the relationship between soil nutrients and crop labels effectively.

4.3 Results of Crop Recommendation

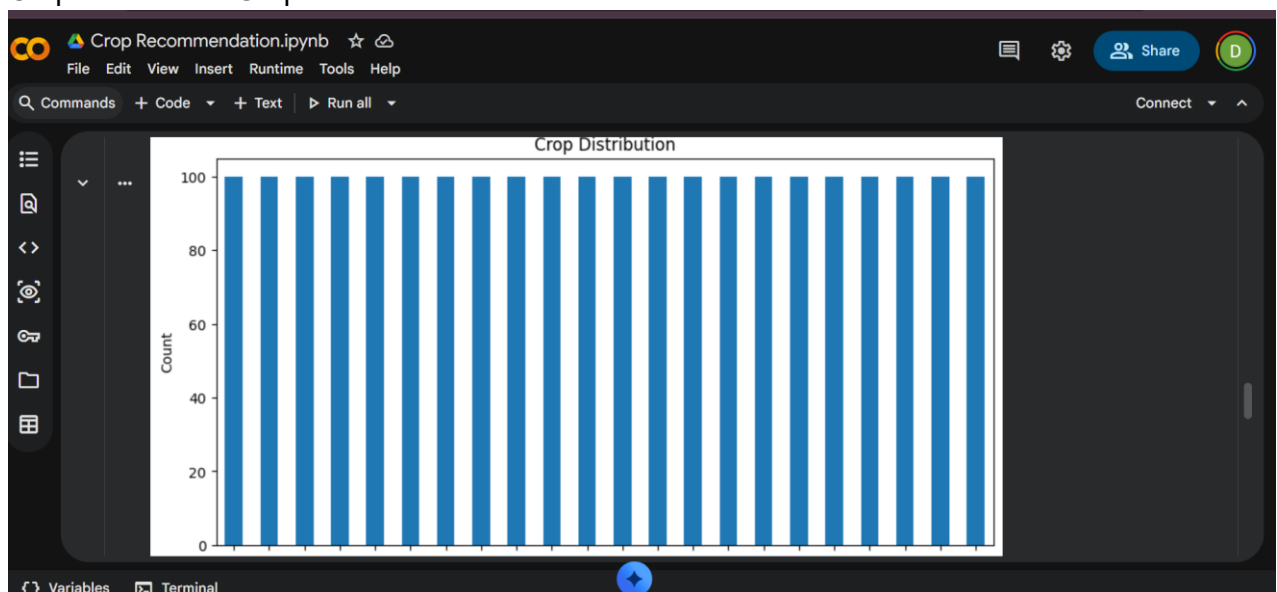
The developed system successfully predicts suitable crops based on user inputs such as Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall.

Performance Results

- Algorithm Used: Random Forest Classifier
 - Number of Trees: 100
 - Dataset Size: 2200 Records
 - Accuracy Achieved: 99.32%

Visualization Results

1. Crop Distribution Graph



2. Correlation Heatmap



3. Prediction Output Screen

```

import pandas as pd

sample = pd.DataFrame(
    [[90, 42, 43, 20.8, 82, 6.5, 202]],
    columns=['N', 'P', 'K', 'temperature', 'humidity', 'ph', 'rainfall']
)

prediction = model.predict(sample)

print("Recommended Crop:", prediction[0])

... Recommended Crop: rice

import pandas as pd

N = float(input("Enter Nitrogen (N): "))
P = float(input("Enter Phosphorus (P): "))
K = float(input("Enter Potassium (K): "))
temperature = float(input("Enter Temperature: "))
humidity = float(input("Enter Humidity: "))
ph = float(input("Enter pH: "))
  
```

The results demonstrate that the proposed model provides highly accurate crop recommendations and can support precision agriculture practices.

CHAPTER 5

DISCUSSION AND CONCLUSION

5.1 Key Findings

1. Machine Learning can effectively recommend crops based on soil and environmental conditions.
2. Random Forest Classifier achieved an accuracy of 99.32%.
3. The system provides reliable recommendations for farmers.
4. Data visualization helps understand agricultural data patterns.

5.2 GitHub Link of the Project

GitHub Repository:

<https://github.com/TammireddyDeepika/AI-Powered-Crop-Recommendation-System-Using-Machine-Learning>

5.3 Video Recording of Project Demonstration

Upload your project demonstration video to Google Drive or YouTube and paste the link here.

5.4 Limitations

1. Depends on the quality of input data.
2. Market demand factors are not considered.
3. Real-time weather forecasting is not integrated.
4. Dataset is limited to available crop records.

5.5 Future Work

1. Integrate real-time weather APIs.
2. Include fertilizer recommendation features.
3. Develop a mobile application for farmers.
4. Expand the dataset for multiple regions.
5. Support multilingual interfaces.

5.6 Conclusion

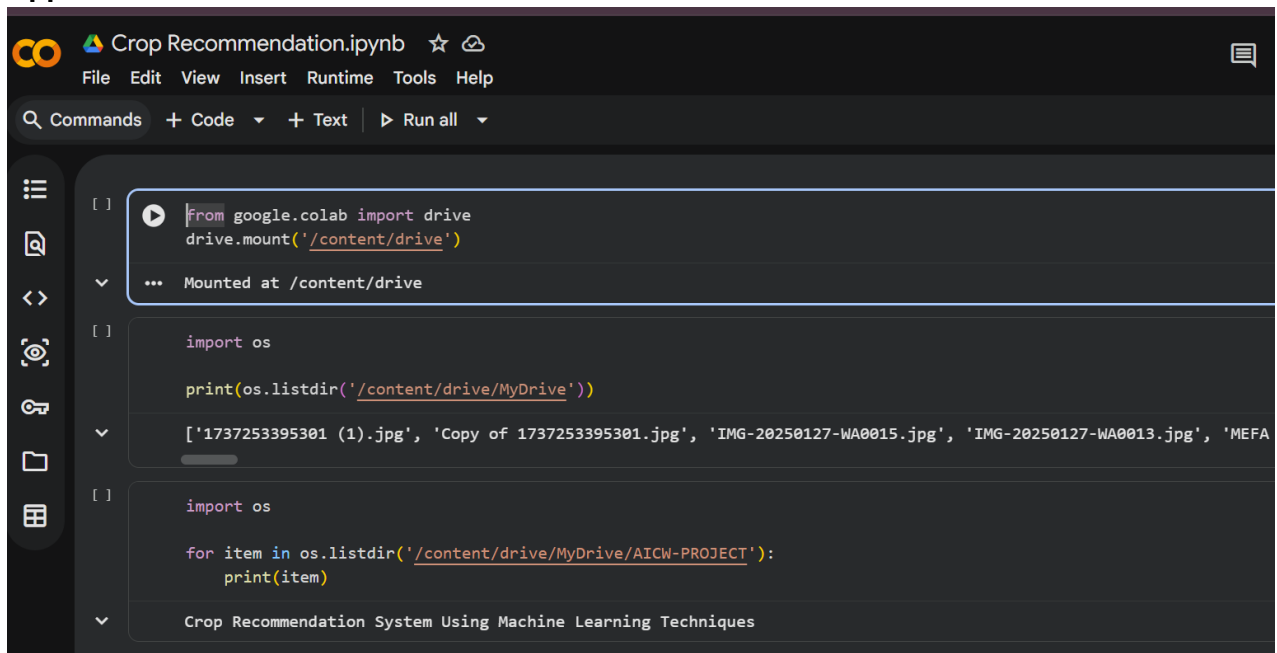
The AI-Powered Crop Recommendation System using Machine Learning successfully recommends suitable crops based on soil nutrients and environmental conditions. The Random Forest model achieved an accuracy of 99.32%, proving its effectiveness in agricultural decision-making. The system can help farmers improve productivity, reduce crop selection risks, and support sustainable farming practices.

REFERENCES

- [1] Crop Recommendation Dataset – Kaggle
- [2] Python Documentation
- [3] Scikit-learn Documentation
- [4] Random Forest Classifier Documentation
- [5] Google Colab Documentation
- [6] GitHub Repository of the Project

APPENDICES

Appendix A – Source Code



```
from google.colab import drive
drive.mount('/content/drive')

... Mounted at /content/drive

import os

print(os.listdir('/content/drive/MyDrive'))

['1737253395301 (1).jpg', 'Copy of 1737253395301.jpg', 'IMG-20250127-WA0015.jpg', 'IMG-20250127-WA0013.jpg', 'MEFA...']

import os

for item in os.listdir('/content/drive/MyDrive/AICW-PROJECT'):
    print(item)

Crop Recommendation System Using Machine Learning Techniques
```

Include Python implementation code.

Appendix B – Dataset Sample

	A	B	C	D	E	F	G	H	
1	N	P	K	temperatu	humidity	ph	rainfall	label	
2	90	42	43	20.87974	82.00274	6.502985	202.9355	rice	
3	85	58	41	21.77046	80.31964	7.038096	226.6555	rice	
4	60	55	44	23.00446	82.32076	7.840207	263.9642	rice	
5	74	35	40	26.4911	80.15836	6.980401	242.864	rice	
6	78	42	42	20.13017	81.60487	7.628473	262.7173	rice	
7	69	37	42	23.05805	83.37012	7.073454	251.055	rice	
8	69	55	38	22.70884	82.63941	5.700806	271.3249	rice	
9	94	53	40	20.27774	82.89409	5.718627	241.9742	rice	
10	89	54	38	24.51588	83.53522	6.685346	230.4462	rice	
11	68	58	38	23.22397	83.03323	6.336254	221.2092	rice	
12	91	53	40	26.52724	81.41754	5.386168	264.6149	rice	
13	90	46	42	23.97898	81.45062	7.502834	250.0832	rice	
14	78	58	44	26.8008	80.88685	5.108682	284.4365	rice	
15	93	56	36	24.01498	82.05687	6.984354	185.2773	rice	
16	94	50	37	25.66585	80.66385	6.94802	209.587	rice	
17	60	48	39	24.28209	80.30026	7.042299	231.0863	rice	
18	85	38	41	21.58712	82.78837	6.249051	276.6552	rice	

< > Crop_recommendation_dataset +

Include sample records from the Crop Recommendation Dataset.

Appendix C – Output Screenshots

The screenshot shows a Jupyter Notebook titled "Crop Recommendation.ipynb". The interface includes a menu bar (File, Edit, View, Insert, Runtime, Tools, Help) and a toolbar with icons for commands, code, text, and running all cells. The notebook contains two code cells. The first cell defines a Random Forest Classifier model, trains it on X_train and y_train, predicts on X_test, and prints the accuracy score. The second cell creates a sample input array and uses the trained model to predict the crop for that sample.

```
[ ]
# Train Model
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy =", accuracy * 100)

... Accuracy = 99.31818181818181

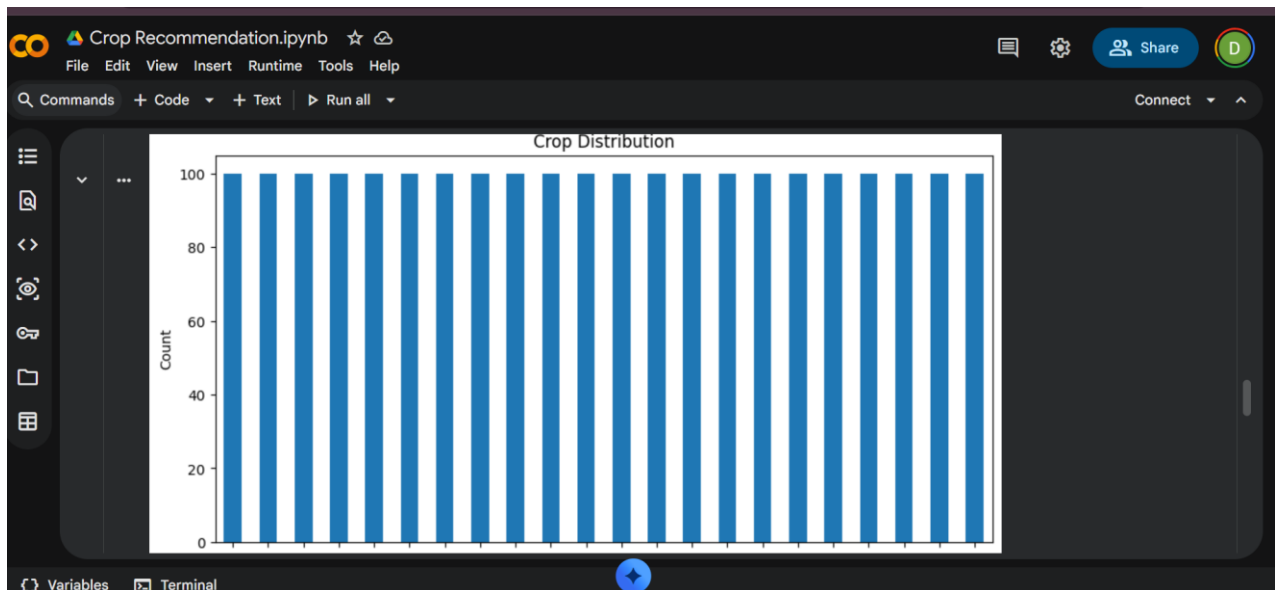
[ ]

sample = [[90, 42, 43, 20.8, 82, 6.5, 202]]

prediction = model.predict(sample)

print("Recommended Crop:", prediction[0])
```

Include Crop Distribution Graph, Correlation Heatmap, and Crop Prediction Screenshots.



Crop Recommendation.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Connect

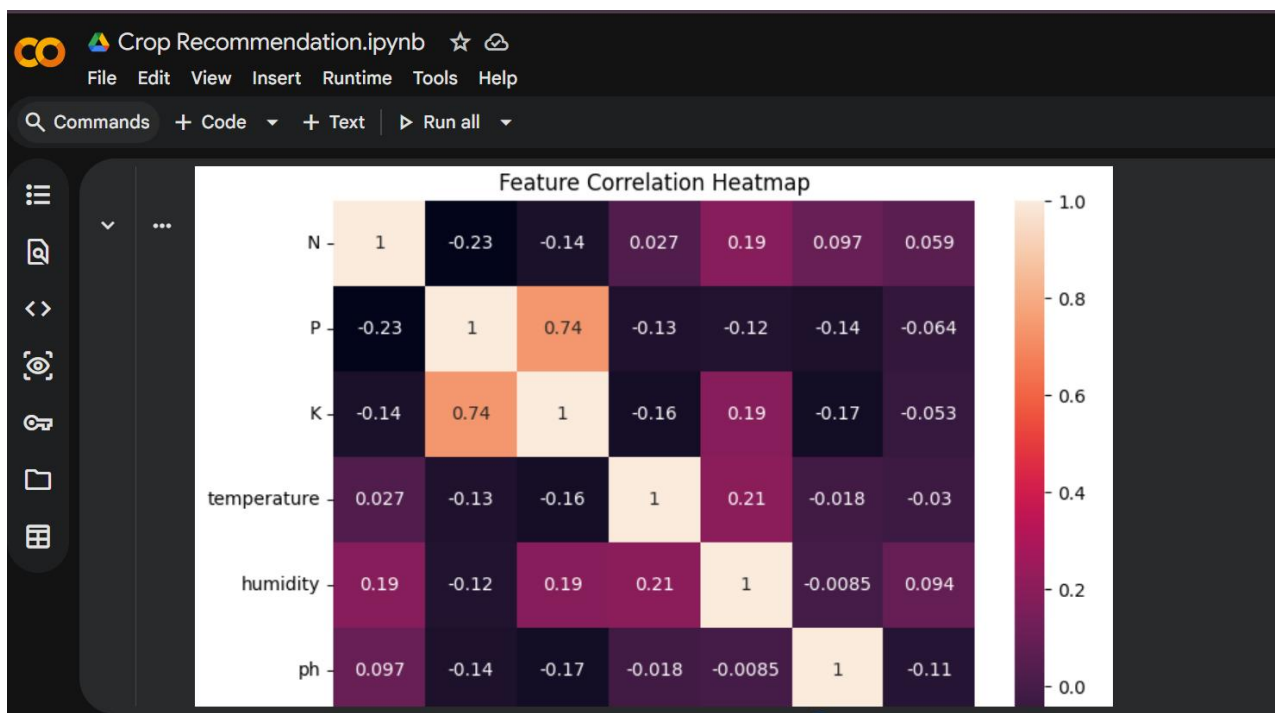
```
[ ] import pandas as pd

df = pd.read_csv('/content/drive/MyDrive/AICW-PROJECT/Crop Recommendation System Using Machine Learning Techniques/Crop_recommendation_dats')

print(df.head())
print(df.columns)
print(df.shape)
```

	N	P	K	temperature	humidity	ph	rainfall	label
0	90	42	43	20.879744	82.002744	6.502985	202.935536	rice
1	85	58	41	21.770462	80.319644	7.038096	226.655537	rice
2	60	55	44	23.004459	82.320763	7.840207	263.964248	rice
3	74	35	40	26.491096	80.158363	6.980401	242.864034	rice
4	78	42	42	20.130175	81.604873	7.628473	262.717340	rice

Index(['N', 'P', 'K', 'temperature', 'humidity', 'ph', 'rainfall', 'label'], dtype='object')
(2200, 8)



 Crop Recommendation.ipynb ☆ ☁

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import pandas as pd

sample = pd.DataFrame(
 [[90, 42, 43, 20.8, 82, 6.5, 202]],
 columns=['N', 'P', 'K', 'temperature', 'humidity', 'ph', 'rainfall']
)

prediction = model.predict(sample)

print("Recommended Crop:", prediction[0])

⌵ ... Recommended Crop: rice

[]

import pandas as pd

N = float(input("Enter Nitrogen (N): "))

P = float(input("Enter Phosphorus (P): "))

K = float(input("Enter Potassium (K): "))

temperature = float(input("Enter Temperature: "))

humidity = float(input("Enter Humidity: "))

ph = float(input("Enter pH: "))

...